

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Supplementary Winter Examination – 2024					
Course: B.Tech.		Branch : Petrochemical Engineering		Semester : VI	
Subject Code & Name: BTPCC604_Y21 Petrochemical Engineering-III					
Max Marks: 60		Date:26/12/2024		Duration: 3 Hr.	
<i>Instructions to the Students:</i> 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.					
				(Level / CO)	Marks
Q.1	Objective type questions. (Compulsory Question)				
1	Molybdenum is used in many steel alloys to_____			Understand	
	a. to increase creep resistance	b. to increase high temperature strength	c. to increase resistance to scaling from oxidation	d. none of these	
2	The selection of tube material in pipe still heater is on the basis of_____ .			Understand	1
	a. corrosion characteristics of process fluid	b. high temperature resistance of material	c. tube metal temperature that governs creep resistance of the material	d. all of these	
3	As a thumb rule , Schedule number of pipe is an indicative of_____.			Understand	1
	a. diameter of pipe	b. thickness of pipe	c. the approx. max pressure that pipe can withstand	d. all of the above	
4	100 percent distilled line on superatmospheric phase diagram represent _____ condition of the stock.			Understand	1
	a. bubble point	b. dew point	c. both (a) and (b)	d. none of the mentioned	
5	Of the below mentioned methods for determining vaporization characteristic of given fraction, which one achieves least fractionation?			Understand	1
	a. ASTM distillation	b. TBP distillation	c. EFV distillation	d. None of these	
6	K value, equilibrium constant, more than one at given condition represent _____distribution of that species in vapor phase than that in liquid phase.			Apply	1
	a. less	b. more	c. equal	d. can't predict	
7	Value of q = 1 for feed entering a distillation column indicates that feed is at its			Understand	1
	a. dew point	b. bubble point	c. critical point	d. none of these	

8	Dehydration of natural gas or petroleum fraction like naphtha and gasoline is carried out by using _____				Remember	1
	a. DEG	b. MEA	c. Lithum Chloride	d. both (a) and (c)		
9	To increase smoke point of kerosene, the compounds removed in post treatment are _____				Understand	1
	a. olefins	b. naphthenes	c. aromatics	d. paraffins		
10	Distillation column operating at lower value than that of minimum reflux ratio will require _____ number of stages for given separation.				Analyze	1
	a. infinite	b. finite	c. minium	d. unpredictable		
11	Doctor process is used with reference to the post treatment of _____				Remember	1
	a. kerosene	b. gasoline	c. diesel	d.LPG		
12	Which of these is not a inter-conversion method for vaporization characteristics determination				Understand	1
	a. Packie Method	b. Underwood Method	c. Edmistser Method	d. Nelson Method		
Q.2	Solve the following.					12
A)	A multicomponent hydrocarbon mixture has the following composition on mole basis. i-butane - 0.20 , n-pentane - 0.25 and n-hexane – 0.55. Calculate dew point temperature and bubble point temperature for this mixture at 100 psia.				Apply	6
B)	In case of flash distillation, how do you calculate the flash? Derive the necessary expression.				Understand	6
Q.3	Solve the following.					12
A)	For a particular stock atmospheric ASTM-D158 distillation data is as follows: Volume % distilled 0 10 30 50 70 90 100 Temp °F 90 140 245 342 408 514 597 Obtain the atmospheric EFV data by using Van winkle method.				Apply	6
B)	Outline method for construction of phase diagram in super-atmospheric region with ASTM D-86 data available at the beginning.				Apply	6
Q.4	Solve Any Two of the following.					12
A)	Draw schematic representation of fractional distillation process. State the minimum essentials for the fractionation process.				Understand	6
B)	Write critically on : Pressure - An important design variable for distillation column.				Analyze	6

C)	Write an informative account on : Limits of operability of distillation column based on reflux ratio.				Analyze	6	
Q.5	Solve Any Two of the following.					12	
A)	Write an informative account on overhead corrosion in distillation unit.				Remember	6	
B)	A quaternary mixture at its dew point composed of component a, b, c and d is to be fractionated to produce the products indicated. The system is to be considered essentially ideal. Calculate the minimum reflux by Underwood method. The reflux is a bubble point liquid.				Apply	6	
	Component	x_f	α	x_D			x_B
	a	0.047	4.2	0.1263			0.0000
	b (LK)	0.072	1.57	0.1913			0.0010
	c (HK)	0.600	1.0	0.6824			0.0350
	d	0.281	0.8	0.0000	0.9649		
C)	Discuss the different aspect with respect to the design of vacuum distillation unit.				Understand	6	
Q.6	Solve Any Two of the following.					12	
A)	Why post treatment of different fractions is necessary? Briefly elaborate on chemical impurities.				Understand	6	
B)	With neat sketch of box type of pipestill heater , give the general description of pipe still heaters.				Remember	6	
C)	Discuss in details the different factors to be considered for design of radiant section of pipe still heater.				Analyze	6	
	*** End ***						